2.10.54

EE25BTECH11057 - Rushil Shanmukha Srinivas

Problem: Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be unit vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$ .Which of the following are correct?

1) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} = \mathbf{0}$
2) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0}$
3) $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{c} \neq \mathbf{0}$
4) $\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}$ are mutually perpendicular.

Solution: Given ,

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}, ||\mathbf{a}|| = ||\mathbf{b}|| = ||\mathbf{c}|| = 1 \tag{4.1}$$

$$(\mathbf{a} + \mathbf{b} + \mathbf{c})^2 = 0 \tag{4.2}$$

$$||\mathbf{a}||^2 + ||\mathbf{b}||^2 + ||\mathbf{c}||^2 + 2\mathbf{a}^\top\mathbf{b} + 2\mathbf{b}^\top\mathbf{c} + 2\mathbf{c}^\top\mathbf{a} = 0 \tag{4.3}$$

$$\mathbf{a}^\top\mathbf{b} + \mathbf{b}^\top\mathbf{c} + \mathbf{c}^\top\mathbf{a} = \frac{-3}{2} \tag{4.4}$$

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0} \implies \begin{pmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0} \implies \mathbf{A}\mathbf{x} = \mathbf{0} \tag{4.5}$$

$$\mathbf{A}^\top\mathbf{A}\mathbf{x} = \mathbf{0} \implies G\mathbf{x} = \mathbf{0} \tag{4.6}$$

$$\begin{pmatrix} ||\mathbf{a}||^2 & \mathbf{a}^\top\mathbf{b} & \mathbf{c}^\top\mathbf{a} \\ \mathbf{a}^\top\mathbf{b} & ||\mathbf{b}||^2 & \mathbf{b}^\top\mathbf{c} \\ \mathbf{c}^\top\mathbf{a} & \mathbf{b}^\top\mathbf{c} & ||\mathbf{c}||^2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0} \tag{4.7}$$

$$\begin{pmatrix} 1 + \mathbf{a}^\top\mathbf{b} + \mathbf{c}^\top\mathbf{a} \\ \mathbf{a}^\top\mathbf{b} + 1 + \mathbf{b}^\top\mathbf{c} \\ \mathbf{c}^\top\mathbf{a} + \mathbf{b}^\top\mathbf{c} + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{4.8}$$

$$1 + \mathbf{a}^\top\mathbf{b} + \mathbf{c}^\top\mathbf{a} = \mathbf{a}^\top\mathbf{b} + 1 + \mathbf{b}^\top\mathbf{c} = \mathbf{c}^\top\mathbf{a} + \mathbf{b}^\top\mathbf{c} + 1 \tag{4.9}$$

$$\mathbf{a}^\top\mathbf{b} = \mathbf{b}^\top\mathbf{c} = \mathbf{c}^\top\mathbf{a} \tag{4.10}$$

From (4.4)

$$\mathbf{a}^\top\mathbf{b} = \mathbf{b}^\top\mathbf{c} = \mathbf{c}^\top\mathbf{a} = \frac{-1}{2} \tag{4.11}$$

If

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} \implies \mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \tag{4.12}$$

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0} \tag{4.13}$$

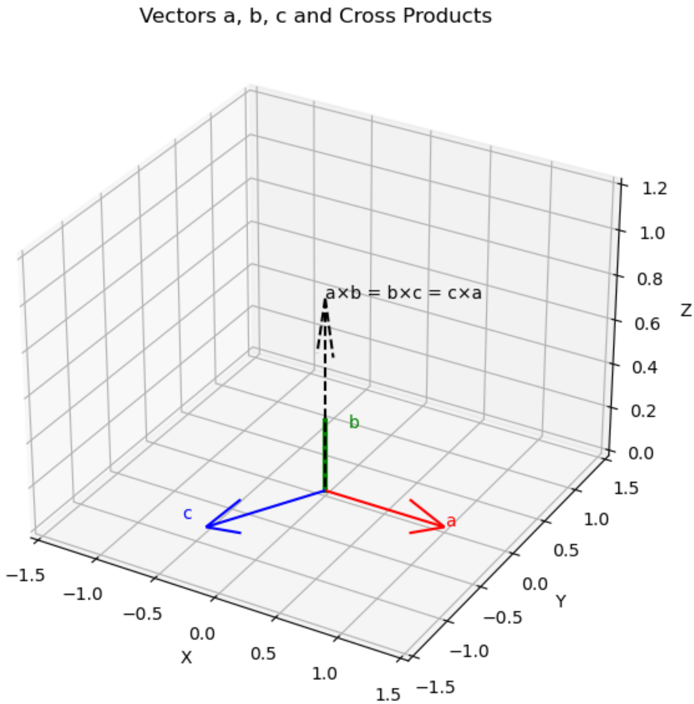

Fig: Representation of vectors